



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (general handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No : BIO F312
Course Title : Plant Physiology
Instructor-In charge : JITENDRA PANWAR (Email: jpanwar@pilani.bits-pilani.ac.in; Room: 3277X)
Instructor : P. Yadukrishnan (Email: yadukrishnan.p@pilani.bits-pilani.ac.in; Room: 3277K)
Prathum Pathak (Research Scholar)

1. Course Description:

Basic functional processes in plants; Plant tissue system, Plant-water relations, Gaseous exchange, Stomatal regulations, Mineral nutrition and absorption, Transport of material, Growth and development, Hormones and Plant Growth Regulators, Photoperiodism, Vernalization, Stress Physiology and related lab components.

2. Scope & Objective:

This course attempts to make the students aware of the major features of physiological aspects of plant system. Emphasis will be given to understand the plant tissue systems, water and mineral nutrient uptake and their distribution, mechanism of transpiration and photosynthate transport, the role of hormones in plant growth & developmental and mechanism of flowering.

3. Text Book:

Taiz, L. and Zeiger, E., 2003. Plant Physiology, 3rd Ed., Panima Publishing Corporation, Indian Reprint.

4. Reference books:

Taiz, L. and Zeiger, E., Plant Physiology, 5th Ed., 2010, Sinauer Associate, Sunderland, USA.

Web Topics: <https://learninglink.oup.com/access/taiz7e>

5. Course Plan

Module No.	Lecture/ Tutorial Session	Reference	Learning Outcome
1. Introduction to the subject and course layout	L 1. Orientation of the course, Introduction to plant physiology	Lecture slides	Familiarization with the plant physiology and diversity
2. Overview of Plant Structure	L 2-4. Plant tissue system (meristematic and permanent); Simple (Parenchyma, Collenchyma, Sclerenchyma) and Complex (xylem, phloem) tissues	Class Notes/ Web topic 1.4	In-depth understanding of various tissue systems, their composition and functions in plants
3. Learning water and its interaction with plant body	L 5-6. Water potential	TB-3, RB-3	Knowledge about the water transport processes and the ascent of sap in plant system
4. How plants obtain water from the soil?	L 7-10. Water movement in soil, absorption by roots and transport through xylem	TB-4, RB-4	
5. The transpiration / photosynthesis paradox	L 11-13. Transpiration and its compromise with photosynthesis, Stomatal regulations, Transpiration ratio	TB-4, RB-4	



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6. Mineral requirements for plant growth	L 14. Mineral Nutrition: Essential elements and their function	TB-5, RB-5	Knowledge about nutrient elements required for plant growth and their deficiency symptoms
7. How are nutrients absorbed and distributed in tissues	L 15-17. Mineral Nutrition: Absorption of minerals, CEC of soils, Mycorrhizal fungi	TB-5, RB-5	Processes involved in nutrient absorption in plants
8. How are nutrients absorbed and distributed in tissues	L 18-20. Transport of solutes and ions, Membrane transport processes, Co-transport and counter transport, Ion trafficking in roots	TB-6, RB-6	Passive and active transport mechanism
9. How are metabolic products distributed in plants?	L 21-23. Transport of material in phloem	TB-10, RB-10	Understanding of pattern and mechanism of phloem translocation, phloem loading and unloading
10. How do plants grow?	L 24-27. Growth and development: Primary & secondary meristem, SAM & RAM, Phyllotaxy	TB-16, RB-16	Understanding of the growth & development of plant shoot, root and leaves
11. What controls plant growth?	L 28-32. Hormones and Growth regulators: Auxins and Gibberellins	TB-19, 20 RB-19, 20	Knowledge about the physiological & developmental effects of PGRs
12. What controls plant growth?	L 33-36. Cytokinins, Absciscic acid, Other important growth regulators	TB-21, 23 RB-21, 23	
13. How plants control the timing of flowering?	L 37-40. Photoreceptors, Photoperiodism, Vernalization, Floral development	TB-17, 24 RB-17, 24	Understanding the effect of light and temperature to control flowering

Practical Assignments:

- 1) Transverse section and observing various tissue systems in dicot shoot & root
- 2) Transverse section and observing various tissue systems in monocot shoot & root
- 3) Transverse section and observing various tissue systems in dicot & monocot leaf

6. Evaluation scheme:

S. No.	Evaluation Component*	Weightage (%)	Date & Time	Remarks (Close Book/ Open Book)
1.	Mid-Semester Test	30	07/10; AN1	CB
2.	Quizzes / Assignments	15	-	CB; Best N-2 out of N will be considered
3.	Practical Assignments	10	-	OB
4.	Comprehensive Exam	45	10/12; FN	CB+OB

*Material in soft-copy will not be allowed in any evaluation component. For Open-Book Exam you are allowed to bring only the indicated books and your own handwritten notes. Printouts/xerox of lecture slides and class notes will NOT be allowed.

Chamber consultation hour: To be announced in the class.

Notices: All notices will be sent on your BITS email address.

Make-up policy: Make-up decisions will be made on a case-by-case basis and only genuine cases as determined by the Departmental team will be considered. No make-up for Quizzes and Practical Assignments.

Note: It shall be the responsibility of the individual student to be regular in attending lectures as per the schedule.

Instructor-in-Charge (BIO F312)